

# Wi-Fi-Controlled ROBOT TOY CAR Using IndusBoard Coin

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This Arduino cloud-based IoT toy car can be controlled remotely using a smartphone, tablet, or laptop connected to the internet. The design can also be extended for applications such as sensor data collection and cloud-based monitoring. Built around the IndusBoard Coin, the system is both compact and lightweight.

The Arduino IoT Cloud acts as the wireless remote-control platform, enabling commands to be transmitted and received over the internet. Fig. 1 shows the author's prototype. The car is driven by N20 DC motors through an L293D motor driver

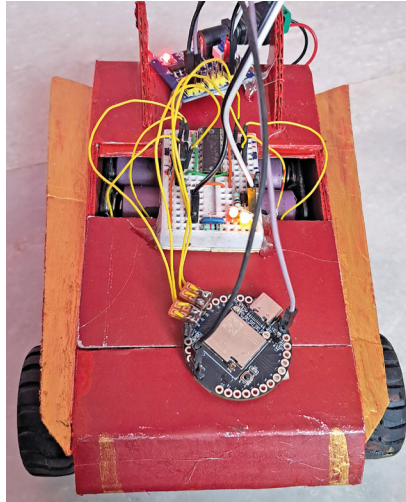
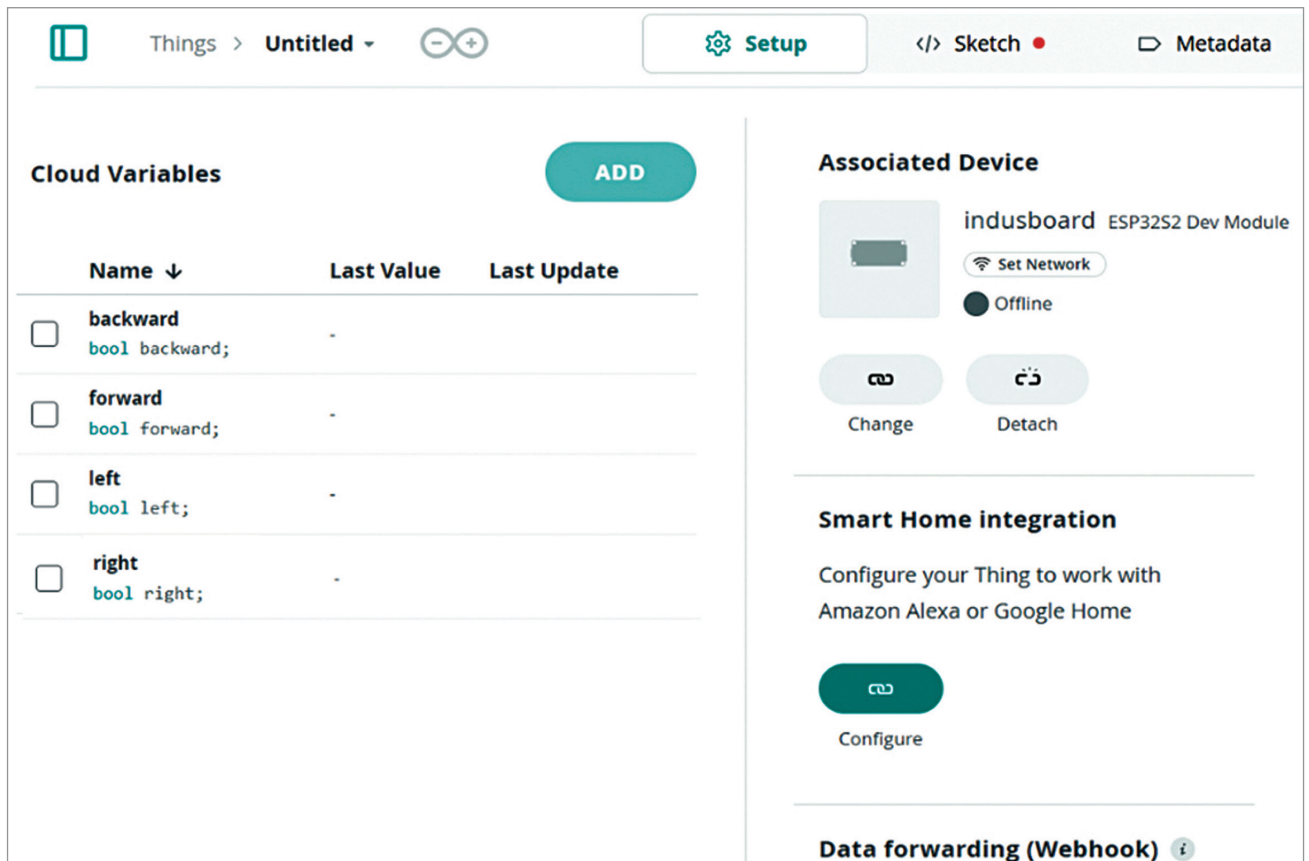


Fig. 1: Author's prototype

IC, while the IndusBoard Coin connects to the Arduino IoT Cloud to receive, process, and execute control commands, driving the motors accordingly. The components required to build the system are listed in the Bill of Materials in Table 1.

## Prerequisite configuration

Create an account on Arduino IoT Cloud and sign in. Then create a new Thing, configure the required variables, generate and edit the sketch, and upload it to the board. Next, create a dashboard and add the necessary widgets to monitor



**Cloud Variables**

|                          | Name ↓                            | Last Value | Last Update |
|--------------------------|-----------------------------------|------------|-------------|
| <input type="checkbox"/> | <b>backward</b><br>bool backward; | -          |             |
| <input type="checkbox"/> | <b>forward</b><br>bool forward;   | -          |             |
| <input type="checkbox"/> | <b>left</b><br>bool left;         | -          |             |
| <input type="checkbox"/> | <b>right</b><br>bool right;       | -          |             |

**Associated Device**

indusboard ESP32S2 Dev Module

Set Network

Offline

Change Detach

**Smart Home integration**

Configure your Thing to work with Amazon Alexa or Google Home

Configure

**Data forwarding (Webhook)**

Fig. 2: Arduino IoT cloud configuration